

Course Code	Course name	L	T	P	C
CSEG4034P	Computational Linguistics and Natural Language Processing	4	0	0	4
Total Units to be Covered: 05		Total Contact Hours: 60			
Prerequisite(s):	Applied Machine Learning - CSAI2016P	Syllabus version: 1.0			

Course Objectives

1. The objectives of this course are:
2. To introduce the concept of Natural Language Understanding & Natural Language Generation.
3. To develop the concept of statistical and probabilistic approach of language modelling.
4. To extend the knowledge of Large Language Model.
5. To enrich the knowledge with different corpuses and different tools being used for machine translation.
6. To provide programming skills necessary for processing natural language.

Course Outcomes

On completion of this course, the students will be able to

CO1: Understand the techniques in NLP.

CO2: To understand the Large Language Model

CO3: To comprehend the natural language generation.

CO4: To understand the machine translation & information retrieval techniques.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	1	-	-	1	1	-	-	-	-	-	-	-	-	-	3
CO 2	1	-	-	1	1	-	-	-	-	-	-	-	-	-	3
CO 3	1	2	-	2	1	-	-	-	-	-	-	-	-	-	3
CO4	1	-	-	3	1	-	-	-	-	-	-	-	-	-	3

Average	1	.5	-	1.75	1	-	-	-	-	-	-	-	-	-	3
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1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

“ - ” means there is no correlation

Syllabus

Unit I: Classical Approaches of NLP

12 Lecture Hours

Introduction, Classical approaches to natural language processing , Understanding linguistics, Level 1: Morphology, Level 2: Syntax, Level 3: Semantics, Level 4: Pragmatics, Understanding linguistics, Traditional approach, Example: Automatic summarization using NLP, Drawbacks, Text processing, Ambiguities and computational challenges in processing various natural languages, Introduction to Real life applications of NLP such as spell and grammar checkers, information extraction, question answering, and machine translation.

Unit II: Empirical & Statistical Approaches

12 Lecture Hours

Corpus creation, Corpus linguistics, Types of corpora, Lexicographical implementations in corpora, Timeline of corpus linguistics, Usage areas of corpora, Traits of a good text corpus, Annotations in text corpus, NLP task-specific training corpora, Treebank annotation, Usage of annotations and corpora, Kinds of annotations, Tree banks and its construction, Need for tree bank, Types of tree bank corpus, Ambiguity in language, Segmentation, Stemming, Tokenization, Representation of word, Sentence, Word embedding, Word Senses, Linguistic Structure: Dependency Parsing. Fundamental statistical techniques, Problems of the traditional approach, how statistics helps, Problems of the traditional approach and how statistics helps, Hidden Markov model, Maximum entropy Markov model, Conditional random field model, Support vector machine, N-GRAM, Perplexity, POS Tagging, Word sense disambiguation, POS tag and Hidden Markov model, POS tagging using HMM, Viterbi algorithm, Recurrent Neural network, Vanishing Gradients and exploding gradients. Parsing, Statistical parsing, Approaches to parsing, Statistical approach, Lexicalized statistical parsing, Top-down parsing, Bottom-up parsing, Left corner parsing method, Statistical parsing: Probabilistic parser,

Unit III: Language Modelling

12 Lecture Hours

Word similarity and text similarity, Text similarity methods, Jaccard similarity, K-means, Cosine similarity, Word Mover's distance, Word sense disambiguation, Complications in WSD, Methods in WSD, Evaluation of WSD, the role of language models, estimating parameters and smoothing. Evaluating language models, LSTM

(Long short-term memory), GRU (Gated recurrent Unit), Part of speech tagging, BERT, XLnet, 1D-CNN for NLP, Sub-word Models, Contextual Representations, Transformers, Self-Attention for Generative Models.

Unit IV: Machine Translation

12 Lecture Hours

Machine translation, Rule-based machine translation, Statically Machine Translation, Neural Machine Translation, Seq2Seq Modelling, Attention, Question Answering Bot, Natural Language Generation, Neural Machine Translation, Case studies on Amazon Alexa, Google Assistant, Microsoft Cortona etc.

Unit V: Applications of Natural Language Processing

12 Lecture Hours

Text Summarization, Document Summarization, Sentiment Analysis, Question Answering system, Sarcasm Detection, Hostile detection Information retrieval in NLP, Image caption generation, Intelligent Tutoring System

Total lecture Hours 60

Textbooks

1. Speech and Language Processing “An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition” Second Edition Daniel Jurafsky Stanford University James H. Martin University of Colorado at Boulder

Reference Books

1. “Information Retrieval: Algorithms and Heuristics” by David A. Grossman, Ophir Frieder, Academic Press.
2. Natural Language Processing A paninian perspective, Prentice Hall, New Delhi, 1994 Akshar Bharati, Vineet Chaitanya, and Rajeev Sangal.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100



Course Code	Course name	L	T	P	C
CSEG4134P	Computational Linguistics and Natural Language Processing Lab	0	0	2	1
Total Units to be Covered: 10		Total Contact Hours: 30			
Prerequisite(s):	Applied Machine Learning Lab - CSAI2016P	Syllabus version: 1.0			

Course Objectives

The objectives of this course are:

1. To introduce the concept of Natural Language Understanding & Natural Language Generation.
2. To develop the concept of statistical and probabilistic approach of language modelling.
3. To extend the knowledge of Large Language Model.
4. To enrich the knowledge with different corpuses and different tools being used for machine translation.
5. To provide programming skills necessary for processing natural language.

Course Outcomes

On completion of this course, the students will be able to

CO1: Understand the techniques in NLP.

CO2: To understand and execute the Large Language Model

CO3: To comprehend and execute the natural language generation.

CO4: To implement the machine translation for various application.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	1	-	-	1	1	-	-	-	-	-	-	-	-	-	3
CO 2	1	-	-	1	1	-	-	-	-	-	-	-	-	-	3
CO 3	1	2	-	2	1	-	-	-	-	-	-	-	-	-	3
CO4	1	-	-	3	1	-	-	-	-	-	-	-	-	-	3
Average	1	.5	-	1.75	1	-	-	-	-	-	-	-	-	-	3

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List of Experiments

EXPERIMENT-1:

Title: Installing various packages required for analytics in python and write code in Python for following program

1. Write code to load CSV file containing information about employee of a company in python and draw graph showing average salary department wise.

EXPERIMENT-2:

Title: Text Retrieval

- 1- Connect to Twitter account and Extract first 100 tweets from it in a file.
- 2- Study and Implementation of Processing text (Word and Sentence Tokenization)

EXPERIMENT-3:

Title: Processing Data

1. Python code to read a text document and perform basic pre-processing techniques on the text like tokenization, stop-word-removal, lemmatization etc.
2. Study and Implementation of Morphological analysis.

EXPERIMENT-4:

Title: Do text mining on extract data and Accessing text corpus

1. Calculate word count of a given specific document and show top 10 frequent words with their frequency and Create word cloud and show graphically.
2. Study and Implementation of NER (Name Entity Recognition)

EXPERIMENT-5:

Title: POS-Tagging and Tagging and Parsing

1. Categorizing and tagging words in Twitter Data.
2. Study and implementation of POS Tagging and Chunking in a sentence.

EXPERIMENT-6:

Title: Language Processor

1. Implement N-Gram Language Model and Smoothing.

EXPERIMENT-7:

Title: Do sentimental analysis

1. Analysis of Sentiment and Subjectivity
2. Implement sentimental analysis on IMDB Movie Reviews Dataset.

EXPERIMENT-8:

Title: Do Text Summarization

1. Analysing Meaning of Sentences
2. Implement Text Summarization on IMDB Movie Reviews Dataset.

EXPERIMENT-9 & 10:

Title: Mini-Project on NLP

Implement Mini-Project on NLP applications.

Total Lab hours 15

Textbooks

1. Speech and Language Processing “An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition” Second Edition Daniel Jurafsky Stanford University James H. Martin University of Colorado at Boulder

Reference Books

1. “Information Retrieval: Algorithms and Heuristics” by David A. Grossman, Ophir Frieder, Academic Press.
2. Natural Language Processing A paninian perspective, Prentice Hall, New Delhi, 1994 Akshar Bharati, Vineet Chaitanya, and Rajeev Sangal.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme: Continuous Assessment

Components	Quiz & Viva	Performance & Lab Report
Weightage (%)	50 %	50 %